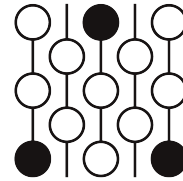


Review of Submission 6B

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

Correctness. The solution is correct. The key inequality, in Lemma 4, is however proven in a very roundabout way, and in particular the “parity estimate” seems mostly to be an avoidable digression. Perhaps the complexity here is coming from the fact that this is an inductive proof of a statement not given in the correct generality: the analogue of this part of the argument in the human solution proves the stronger statement that the desired inequality holds whenever $\lambda_r \neq 0$.

Novelty. This argument provides a correct solution but little more than that – it is difficult to extract the “story” of why the result holds from the given arguments, and in fact seems that there is no broader idea driving the proof. At their core, all of the successful solutions to this problem are an induction down the tree where needed inequalities at a root follow from those same inequalities in subtrees. The loop does close in this solution and lead to a correct proof, but perhaps it’s not a priori clear why. This contrasts with the human proof, which phrases the intermediate inequalities in terms of the geometry of the dual lattice and it’s a bit easier to see a throughline.

2 Recommendation

On a technical level, this is **essentially flawless**. However, the fairly technical arguments are sufficiently interleaved and unmotivated that it is quite difficult to check line-by-line, making it hard to recommend publication without further polish.

3 Submission with inline comments

Overview / Strategy

The proof has two main parts. First, a vertex of weight 1 is automatically irreducible, because a decomposition $u = p + q$ with $p \cdot q \geq 0$ would force $1 = u^2 \geq p^2 + q^2 \geq 2$. Thus, after this easy case, we may assume that no vertex has weight 1.

In that remaining case, let b be the unique vertex with $w(b) < d_T(b)$. We prove that b itself is irreducible. The key input is a rooted-branch estimate for each component of $T - \{b\}$. That

estimate is proved by induction using a Schur-complement calculation and a parity argument for the inverse Gram matrix. Combining the branch estimates with the Schur complement at b , any hypothetical decomposition $b = x + (b - x)$ with $x \cdot (b - x) \geq 0$ is forced to have $x = 0$ or $x = b$, contradicting the requirement that both summands be nonzero.

Statement and notation

Let T be a finite tree with vertex set V , edge set E , and weight function $w : V \rightarrow \mathbb{Z}$. We write $d_T(v)$ for the degree of a vertex v in T . Let $L(T)$ be the free abelian group generated by V . We identify a vertex with the corresponding basis element of $L(T)$. The bilinear form on $L(T)$ is the symmetric form determined on generators by

$$u \cdot v = \begin{cases} w(u), & u = v, \\ -1, & \{u, v\} \in E, \\ 0, & u \neq v \text{ and } \{u, v\} \notin E. \end{cases}$$

For $x \in L(T)$, write $x^2 = x \cdot x$. The form is called positive definite if its extension to $L(T) \otimes_{\mathbb{Z}} \mathbb{R}$ is positive definite. Since the Gram matrix has integer entries, if the form is positive definite, then every nonzero $z \in L(T)$ satisfies $z^2 \in \mathbb{Z}_{>0}$.

A nonzero lattice element $y \in L(T)$ is called *irreducible* if there do not exist nonzero $a, c \in L(T)$ such that

$$y = a + c \quad \text{and} \quad a \cdot c \geq 0.$$

Theorem 1 (Original problem). *Let T be a finite weighted tree with vertex set V , edge set E , and weight function $w : V \rightarrow \mathbb{Z}$. Equip $L(T)$ with the symmetric bilinear form above. Suppose this form is positive definite, and suppose there is exactly one vertex $b \in V$ such that*

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Then T contains a vertex u such that u is irreducible in $L(T)$.

For a connected subgraph or subtree C , write $d_C(u)$ for the degree of u inside C . The group $L(C)$ is the free abelian group on the vertices of C , equipped with the restricted form. If $x \in L(C)$, write x_u for the coefficient of the vertex u in x .

Auxiliary lemmas

Lemma 2 (A weight-one vertex is irreducible). *Assume the form on $L(T)$ is positive definite. If $u \in V$ satisfies $w(u) = 1$, then u is irreducible.*

Proof. Suppose, for contradiction, that

$$u = p + q$$

where $p, q \in L(T)$ are nonzero and $p \cdot q \geq 0$. Since the form is positive definite and integral, $p^2, q^2 \in \mathbb{Z}_{>0}$. Hence

$$1 = u^2 = (p + q)^2 = p^2 + q^2 + 2p \cdot q \geq 1 + 1 + 0 = 2,$$

which is impossible. Therefore no such decomposition exists, and u is irreducible. $\square$

Lemma 3 (Schur complement calculation). *Let*

$$M = \begin{pmatrix} a & \gamma^T \\ \gamma & B \end{pmatrix},$$

where B is symmetric positive definite. Set

$$S = a - \gamma^T B^{-1} \gamma.$$

Then M is positive definite if and only if $S > 0$. When $S > 0$,

$$M^{-1} = \begin{pmatrix} S^{-1} & -S^{-1} \gamma^T B^{-1} \\ -B^{-1} \gamma S^{-1} & B^{-1} + B^{-1} \gamma S^{-1} \gamma^T B^{-1} \end{pmatrix}.$$

Proof. We have the factorization

$$M = \begin{pmatrix} 1 & \gamma^T B^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} S & 0 \\ 0 & B \end{pmatrix} \begin{pmatrix} 1 & 0 \\ B^{-1} \gamma & I \end{pmatrix}.$$

Indeed, multiplying the last two factors gives

$$\begin{pmatrix} S & 0 \\ \gamma & B \end{pmatrix},$$

and multiplying by the first factor gives

$$\begin{pmatrix} S + \gamma^T B^{-1} \gamma & \gamma^T \\ \gamma & B \end{pmatrix} = \begin{pmatrix} a & \gamma^T \\ \gamma & B \end{pmatrix}.$$

The first and third factors are transposes of one another and are invertible. Congruence by an invertible matrix preserves positive definiteness, so M is positive definite exactly when the middle block diagonal matrix is positive definite. Since B is already positive definite, this is equivalent to $S > 0$.

When $S > 0$, invert the displayed factorization:

$$M^{-1} = \begin{pmatrix} 1 & 0 \\ -B^{-1} \gamma & I \end{pmatrix} \begin{pmatrix} S^{-1} & 0 \\ 0 & B^{-1} \end{pmatrix} \begin{pmatrix} 1 & -\gamma^T B^{-1} \\ 0 & I \end{pmatrix}.$$

Multiplying these three factors gives exactly the stated block formula. $\square$

Lemma 4 (Rooted branch estimate). *Let C be a finite weighted tree rooted at a vertex r . Let A be its Gram matrix, and assume A is positive definite. Suppose*

$$w(r) \geq d_C(r) + 1, \quad w(u) \geq d_C(u) \quad \text{for every } u \neq r.$$

Let $G = A^{-1}$, and define

$$\alpha = (A^{-1})_{rr} = e_r^T G e_r,$$

where e_r is the coordinate vector of r . Then $0 < \alpha \leq 1$, and for every $x \in L(C)$, if $c = x_r$, then for every $k \in \mathbb{Z}$,

$$(2k - 1)c - x^2 \leq k(k - 1)\alpha.$$

Moreover, if $k = 0$ or $k = 1$, and equality holds with $x \neq 0$, then C contains a vertex of weight 1.

The statement is correct, but the parity restrictions that appear in the proof seem to be unnecessary and don't add any clarity.

Proof. We first prove a parity estimate for inverse Gram matrices. We claim that, under the hypotheses of the lemma,

$$0 < \alpha \leq 1$$

and that every vector $\lambda \in \mathbb{Z}^C$ satisfying

$$\lambda_r \equiv 1 \pmod{2}, \quad \lambda_u \equiv 0 \pmod{2} \quad (u \neq r)$$

also satisfies

$$\lambda^T G \lambda \geq \alpha.$$

This is proved by induction on $|C|$.

If $C = \{r\}$, then $A = [w(r)]$. The root hypothesis gives $w(r) \geq 1$, and positive definiteness also gives $w(r) > 0$. Thus

$$\alpha = \frac{1}{w(r)} \leq 1.$$

If $\lambda_r = s$ is odd, then $s^2 \geq 1$, so

$$\lambda^T G \lambda = \frac{s^2}{w(r)} \geq \frac{1}{w(r)} = \alpha.$$

This proves the base case.

Assume now that $|C| > 1$. Let $C_1, \dots, C_m$ be the connected components of $C - \{r\}$, and let r_i be the unique vertex of C_i adjacent to r . Root C_i at r_i . Let A_i be the Gram matrix of C_i , set $G_i = A_i^{-1}$, let $e_i = e_{r_i}$, and put

$$\alpha_i = e_i^T G_i e_i.$$

Each A_i is positive definite because it is a principal submatrix of A . Also, each rooted tree C_i satisfies the same type of weight hypotheses: since $r_i \neq r$,

$$w(r_i) \geq d_C(r_i) = d_{C_i}(r_i) + 1,$$

and every other vertex $u \in C_i$ satisfies

$$w(u) \geq d_C(u) = d_{C_i}(u).$$

Therefore, by induction,

$$0 < \alpha_i \leq 1 \quad \text{for all } i.$$

Order the vertices of C by putting r first and then the vertices of $C_1, \dots, C_m$. With respect to this order,

$$A = \begin{pmatrix} w(r) & -\eta^T \\ -\eta & B \end{pmatrix}, \quad B = A_1 \oplus \dots \oplus A_m,$$

where the C_i -block of η is e_i . By Lemma 3, if

$$S = w(r) - \eta^T B^{-1} \eta,$$

then $S > 0$ and

$$\alpha = (A^{-1})_{rr} = \frac{1}{S}.$$

Since $B^{-1} = G_1 \oplus \cdots \oplus G_m$, we have

$$\eta^T B^{-1} \eta = \sum_i \alpha_i.$$

Also $m = d_C(r)$, so the root hypothesis and the inductive bounds $\alpha_i \leq 1$ imply

$$S = w(r) - \sum_i \alpha_i \geq (m+1) - m = 1.$$

Thus $0 < \alpha = 1/S \leq 1$.

Now let $\lambda \in \mathbb{Z}^C$ have odd root coordinate and even nonroot coordinates. In the above ordering, write

$$\lambda = (s, 2\mu_1, \dots, 2\mu_m),$$

where s is odd and $\mu_i \in \mathbb{Z}^{C_i}$. Define

$$\rho_i = e_i^T G_i \mu_i, \quad E_i = \mu_i^T G_i \mu_i, \quad \rho = \sum_i \rho_i.$$

Using the inverse formula from Lemma 3 with $\gamma = -\eta$, we obtain

$$\lambda^T G \lambda = 4 \sum_i E_i + \frac{(s+2\rho)^2}{S}.$$

For each i , the vectors $2\mu_i - e_i$ and $2\mu_i + e_i$ have odd coordinate at the root r_i and even coordinates elsewhere in C_i . The induction hypothesis applied to C_i gives

$$(2\mu_i - e_i)^T G_i (2\mu_i - e_i) \geq \alpha_i, \quad (2\mu_i + e_i)^T G_i (2\mu_i + e_i) \geq \alpha_i.$$

Expanding these two inequalities gives

$$4E_i - 4\rho_i + \alpha_i \geq \alpha_i, \quad 4E_i + 4\rho_i + \alpha_i \geq \alpha_i.$$

Hence

$$E_i \geq |\rho_i|.$$

Therefore

$$\lambda^T G \lambda \geq 4 \sum_i |\rho_i| + \frac{(s+2\rho)^2}{S} \geq 4|\rho| + \frac{(s+2\rho)^2}{S}.$$

Set $y = 2\rho$. Then

$$\lambda^T G \lambda \geq 2|y| + \frac{(s+y)^2}{S}.$$

Since $S \geq 1$, it is enough to prove

$$(s+y)^2 + 2S|y| \geq 1.$$

Let $a = |s| \geq 1$ and $u = |y|$. If $sy \geq 0$, then $|s+y| = a+u \geq 1$, so the inequality is immediate. If $sy < 0$, then

$$(s+y)^2 + 2S|y| \geq (a-u)^2 + 2u = (u - (a-1))^2 + 2a - 1 \geq 1.$$

Thus $S\lambda^T G \lambda \geq 1$, and hence

$$\lambda^T G \lambda \geq \frac{1}{S} = \alpha.$$

This completes the induction and proves the parity estimate.

We now derive the stated branch inequality. Fix $x \in L(C)$, let $c = x_r$, and set $Q = x^2 = x^T A x$. For $k \in \mathbb{Z}$, put

$$t = 2k - 1.$$

Define

$$\lambda = 2Ax - te_r.$$

Because A has integer entries and x has integer coordinates, $\lambda \in \mathbb{Z}^C$. Its root coordinate is even minus the odd integer t , hence is odd, while every nonroot coordinate is even. The parity estimate therefore gives

$$\lambda^T G \lambda \geq \alpha.$$

Using $G = A^{-1}$, we expand:

$$\lambda^T G \lambda = (2Ax - te_r)^T G (2Ax - te_r) = 4Q - 4tc + t^2\alpha.$$

Hence

$$4Q - 4tc + t^2\alpha \geq \alpha,$$

or equivalently

$$tc - Q \leq \frac{t^2 - 1}{4}\alpha.$$

Since $t = 2k - 1$,

$$\frac{t^2 - 1}{4} = \frac{(2k - 1)^2 - 1}{4} = k(k - 1).$$

Therefore

$$(2k - 1)c - x^2 \leq k(k - 1)\alpha.$$

It remains to prove the equality statement for $k = 0, 1$. For every $x \in L(C)$, expanding the form gives

$$x^2 = \sum_{\{u,v\} \in E(C)} (x_u - x_v)^2 + \sum_{u \in C} (w(u) - d_C(u))x_u^2.$$

Indeed, in the first sum, the term x_u^2 appears once for each edge incident to u , hence $d_C(u)$ times, and each edge contributes the cross term $-2x_u x_v$. Therefore the displayed expression equals

$$\sum_{u \in C} w(u) x_u^2 - 2 \sum_{\{u,v\} \in E(C)} x_u x_v = x^2.$$

By the hypotheses,

$$w(r) - d_C(r) \geq 1, \quad w(u) - d_C(u) \geq 0 \quad (u \neq r),$$

so

$$x^2 - c^2 = \sum_{\{u,v\} \in E(C)} (x_u - x_v)^2 + (w(r) - d_C(r) - 1)c^2 + \sum_{u \neq r} (w(u) - d_C(u))x_u^2 \geq 0.$$

Thus $x^2 \geq c^2$.

Suppose first that $k = 1$, equality holds in the branch inequality, and $x \neq 0$. Since $k(k-1) = 0$, equality says

$$c - x^2 = 0.$$

Thus $x^2 = c$. Positive definiteness gives $x^2 > 0$, so $c > 0$, and hence $x^2 = |c|$. Since $x^2 \geq c^2$, we get $|c| \geq c^2$. Because $c \in \mathbb{Z}$ and $c \neq 0$, this forces $c = 1$, and then $x^2 = c^2 = 1$.

Similarly, if $k = 0$, equality says

$$-c - x^2 = 0.$$

Thus $x^2 = -c > 0$, so $c < 0$. Again $x^2 = |c|$, and the inequality $x^2 \geq c^2$ forces $c = -1$ and $x^2 = c^2 = 1$.

In either case, $|c| = 1$ and $x^2 = c^2$. Therefore equality holds in the nonnegative sum for $x^2 - c^2$, so every summand in that sum is zero. In particular, each edge term $(x_u - x_v)^2$ is zero. Since C is connected, all coordinates of x are equal to $c = \pm 1$.

If C has only one vertex, then $C = \{r\}$. The equality $x^2 = c^2$, with $c^2 = 1$, gives

$$(w(r) - d_C(r) - 1)c^2 = 0.$$

Since $d_C(r) = 0$, we obtain $w(r) = 1$.

If C has more than one vertex, choose a vertex $u \neq r$ at maximal distance from r . Such a vertex is a nonroot leaf, so $d_C(u) = 1$. Because all coordinates of x are ± 1 , we have $x_u^2 = 1$. The vanishing of the corresponding nonroot summand gives

$$(w(u) - d_C(u))x_u^2 = 0,$$

hence $w(u) - d_C(u) = 0$, and therefore $w(u) = 1$. Thus, whenever equality holds for $k = 0$ or $k = 1$ with $x \neq 0$, the branch C contains a vertex of weight 1. $\square$

Lemma 5 (Calculation at a vertex). *Let T be a finite weighted tree whose form is positive definite, and let b be a vertex of T . Let C range over the components of $T - \{b\}$. For each such component C , let r_C be the unique vertex of C adjacent to b , let A_C be the Gram matrix of C , and define*

$$\alpha_C = (A_C^{-1})_{r_C r_C}.$$

Then

$$\sum_C \alpha_C < w(b).$$

Moreover, if

$$x = kb + \sum_C x_C,$$

where $k \in \mathbb{Z}$ and $x_C \in L(C)$, and if $c_C = (x_C)_{r_C}$, then

$$x \cdot (b - x) = k(1 - k)w(b) + \sum_C \left((2k - 1)c_C - x_C^2 \right).$$

Proof. If $T = \{b\}$, then there are no components C , the sum $\sum_C \alpha_C$ is empty, and the first assertion says $0 < w(b)$, which follows from positive definiteness because $w(b) = b^2$. The displayed formula also reduces immediately to

$$(kb) \cdot (b - kb) = k(1 - k)w(b).$$

Thus assume $T \neq \{b\}$.

Order the vertices of T by putting b first, followed by the vertices in the components C of $T - \{b\}$. With respect to this order, the full Gram matrix has block form

$$\begin{pmatrix} w(b) & -\eta^T \\ -\eta & B \end{pmatrix}, \quad B = \bigoplus_C A_C,$$

where the C -block of η is e_{r_C} . The matrix B is positive definite because it is a principal submatrix of the positive definite Gram matrix of T . Applying Lemma 3, the Schur complement is positive:

$$w(b) - \eta^T B^{-1} \eta > 0.$$

Since $B^{-1} = \bigoplus_C A_C^{-1}$, we have

$$\eta^T B^{-1} \eta = \sum_C \alpha_C.$$

Therefore

$$\sum_C \alpha_C < w(b).$$

Now write

$$x = kb + \sum_C x_C.$$

Different components of $T - \{b\}$ are orthogonal to one another, and b pairs nontrivially with a component C only at the root r_C , where $b \cdot r_C = -1$. Hence

$$b \cdot x = kw(b) - \sum_C c_C.$$

Also,

$$x^2 = k^2 w(b) - 2k \sum_C c_C + \sum_C x_C^2.$$

Therefore

$$\begin{aligned}
x \cdot (b - x) &= x \cdot b - x^2 \\
&= \left(kw(b) - \sum_C c_C \right) - \left(k^2 w(b) - 2k \sum_C c_C + \sum_C x_C^2 \right) \\
&= k(1 - k)w(b) + \sum_C \left((2k - 1)c_C - x_C^2 \right).
\end{aligned}$$

This proves the formula. □

Proof of the theorem

Proof of Theorem 1. Let b denote the unique vertex of T satisfying $w(b) < d_T(b)$. Thus every vertex $u \neq b$ satisfies

$$w(u) \geq d_T(u).$$

If T contains a vertex of weight 1, then Lemma 2 shows that this vertex is irreducible, and the theorem follows. Hence assume from now on that T contains no vertex of weight 1. We will prove that b is irreducible.

Suppose, for contradiction, that b is reducible. Then there exists $x \in L(T)$ such that

$$x \neq 0, \quad b - x \neq 0, \quad x \cdot (b - x) \geq 0.$$

Decompose x according to the components of $T - \{b\}$:

$$x = kb + \sum_C x_C,$$

where $k \in \mathbb{Z}$, each $x_C \in L(C)$, and the sum ranges over the components C of $T - \{b\}$. Let r_C be the vertex of C adjacent to b , and put

$$c_C = (x_C)_{r_C}.$$

We now verify that each rooted component C , rooted at r_C , satisfies the hypotheses of Lemma 4. Since $r_C \neq b$, the uniqueness of b gives

$$w(r_C) \geq d_T(r_C) = d_C(r_C) + 1,$$

because the only edge incident to r_C that is not in C is the edge from r_C to b . If $u \in C$ and $u \neq r_C$, then no edge incident to u was removed when passing from T to C , so

$$w(u) \geq d_T(u) = d_C(u).$$

Finally, the Gram matrix A_C of C is positive definite because it is a principal submatrix of the positive definite Gram matrix of T . Thus Lemma 4 applies to each component C .

Let

$$\alpha_C = (A_C^{-1})_{r_C r_C}.$$

Applying Lemma 4 to each component, with the same integer k , gives

$$(2k - 1)c_C - x_C^2 \leq k(k - 1)\alpha_C.$$

Using Lemma 5, we obtain

$$\begin{aligned} x \cdot (b - x) &= k(1 - k)w(b) + \sum_C \left((2k - 1)c_C - x_C^2 \right) \\ &\leq k(1 - k)w(b) + k(k - 1) \sum_C \alpha_C \\ &= k(k - 1) \left(\sum_C \alpha_C - w(b) \right). \end{aligned}$$

By Lemma 5,

$$\sum_C \alpha_C - w(b) < 0.$$

If $k \notin \{0, 1\}$, then $k(k - 1) > 0$, so the last upper bound is strictly negative. This contradicts $x \cdot (b - x) \geq 0$. Therefore

$$k \in \{0, 1\}.$$

For $k = 0$ or $k = 1$, we have $k(1 - k) = 0$, and Lemma 4 gives

$$(2k - 1)c_C - x_C^2 \leq 0 \quad \text{for every component } C.$$

Lemma 5 therefore gives

$$x \cdot (b - x) = \sum_C \left((2k - 1)c_C - x_C^2 \right),$$

a sum of nonpositive terms. Since $x \cdot (b - x) \geq 0$, every term in this sum must be equal to 0. Hence, for every component C ,

$$(2k - 1)c_C - x_C^2 = 0.$$

If some $x_C \neq 0$, then the equality statement in Lemma 4 implies that C , and therefore T , contains a vertex of weight 1. This contradicts our standing assumption that T has no weight-one vertex. Thus

$$x_C = 0 \quad \text{for every component } C.$$

Consequently $x = kb$. If $k = 0$, then $x = 0$, contradicting $x \neq 0$. If $k = 1$, then $b - x = 0$, contradicting $b - x \neq 0$. Both possibilities are impossible.

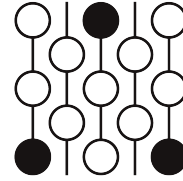
Therefore no such decomposition of b exists, and b is irreducible. Hence T contains an irreducible vertex. $\square$

Review of Submission 6B

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. The proof is correct and pretty clearly explained, although the writing is far from perfect. What I find appealing is that it explains a lot of detail, more than any of the other solutions, which made it easier to check. On the other hand, the conceptual organization can get a little lost in what feels like a stream of consciousness proof.

Novelty. I don't know to what extent this argument takes ideas from the literature. It's not always clear about when it's giving a standard derivation of well-known material, such as the Schur complement calculation in Lemma 3, and when it's doing something original. It's clearly not trying to deceive; for example, it uses the term Schur complement. However, it would be useful to be more explicit.

2 Recommendation

I do not see any mathematical issues that require correction. The writing could use some improvement. For example, I would consider splitting Lemma 4 into two lemmas, and maybe isolating the parity estimate from the beginning of the proof as its own lemma. The writing has a tendency to just power through things stream of consciousness style, in a way that is pretty routine to check but is not the most useful for providing conceptual organization and insight.

With a human author, I would view this level of editing as minor revisions not requiring further review, but I'd be a little more skeptical of AI's ability to carry out the revisions well. In any case, these revisions would not be mathematical corrections.

3 Submission with inline comments

Overview / Strategy

The proof has two main parts. First, a vertex of weight 1 is automatically irreducible, because a decomposition $u = p + q$ with $p \cdot q \geq 0$ would force $1 = u^2 \geq p^2 + q^2 \geq 2$. Thus, after this easy case, we may assume that no vertex has weight 1.

In that remaining case, let b be the unique vertex with $w(b) < d_T(b)$. We prove that b itself is irreducible. The key input is a rooted-branch estimate for each component of $T - \{b\}$. That estimate is proved by induction using a Schur-complement calculation and a parity argument for the inverse Gram matrix. Combining the branch estimates with the Schur complement at b , any hypothetical decomposition $b = x + (b - x)$ with $x \cdot (b - x) \geq 0$ is forced to have $x = 0$ or $x = b$, contradicting the requirement that both summands be nonzero.

Statement and notation

Let T be a finite tree with vertex set V , edge set E , and weight function $w : V \rightarrow \mathbb{Z}$. We write $d_T(v)$ for the degree of a vertex v in T . Let $L(T)$ be the free abelian group generated by V . We identify a vertex with the corresponding basis element of $L(T)$. The bilinear form on $L(T)$ is the symmetric form determined on generators by

$$u \cdot v = \begin{cases} w(u), & u = v, \\ -1, & \{u, v\} \in E, \\ 0, & u \neq v \text{ and } \{u, v\} \notin E. \end{cases}$$

For $x \in L(T)$, write $x^2 = x \cdot x$. The form is called positive definite if its extension to $L(T) \otimes_{\mathbb{Z}} \mathbb{R}$ is positive definite. Since the Gram matrix has integer entries, if the form is positive definite, then every nonzero $z \in L(T)$ satisfies $z^2 \in \mathbb{Z}_{>0}$.

A nonzero lattice element $y \in L(T)$ is called *irreducible* if there do not exist nonzero $a, c \in L(T)$ such that

$$y = a + c \quad \text{and} \quad a \cdot c \geq 0.$$

Theorem 1 (Original problem). *Let T be a finite weighted tree with vertex set V , edge set E , and weight function $w : V \rightarrow \mathbb{Z}$. Equip $L(T)$ with the symmetric bilinear form above. Suppose this form is positive definite, and suppose there is exactly one vertex $b \in V$ such that*

$$w(b) < d_T(b).$$

Then T contains a vertex u such that u is irreducible in $L(T)$.

For a connected subgraph or subtree C , write $d_C(u)$ for the degree of u inside C . The group $L(C)$ is the free abelian group on the vertices of C , equipped with the restricted form. If $x \in L(C)$, write x_u for the coefficient of the vertex u in x .

Auxiliary lemmas

Lemma 2 (A weight-one vertex is irreducible). *Assume the form on $L(T)$ is positive definite. If $u \in V$ satisfies $w(u) = 1$, then u is irreducible.*

Proof. Suppose, for contradiction, that

$$u = p + q$$

where $p, q \in L(T)$ are nonzero and $p \cdot q \geq 0$. Since the form is positive definite and integral, $p^2, q^2 \in \mathbb{Z}_{>0}$. Hence

$$1 = u^2 = (p + q)^2 = p^2 + q^2 + 2p \cdot q \geq 1 + 1 + 0 = 2,$$

which is impossible. Therefore no such decomposition exists, and u is irreducible. $\square$

Lemma 3 (Schur complement calculation). *Let*

$$M = \begin{pmatrix} a & \gamma^T \\ \gamma & B \end{pmatrix},$$

where B is symmetric positive definite. Set

$$S = a - \gamma^T B^{-1} \gamma.$$

Then M is positive definite if and only if $S > 0$. When $S > 0$,

$$M^{-1} = \begin{pmatrix} S^{-1} & -S^{-1} \gamma^T B^{-1} \\ -B^{-1} \gamma S^{-1} & B^{-1} + B^{-1} \gamma S^{-1} \gamma^T B^{-1} \end{pmatrix}.$$

Reviewer 2

It would be good to say Lemma 3 is a standard result and proof, being given here since it may not be familiar to all readers.

Proof. We have the factorization

$$M = \begin{pmatrix} 1 & \gamma^T B^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} S & 0 \\ 0 & B \end{pmatrix} \begin{pmatrix} 1 & 0 \\ B^{-1} \gamma & I \end{pmatrix}.$$

Indeed, multiplying the last two factors gives

$$\begin{pmatrix} S & 0 \\ \gamma & B \end{pmatrix},$$

and multiplying by the first factor gives

$$\begin{pmatrix} S + \gamma^T B^{-1} \gamma & \gamma^T \\ \gamma & B \end{pmatrix} = \begin{pmatrix} a & \gamma^T \\ \gamma & B \end{pmatrix}.$$

The first and third factors are transposes of one another and are invertible. Congruence by an invertible matrix preserves positive definiteness, so M is positive definite exactly when the middle block diagonal matrix is positive definite. Since B is already positive definite, this is equivalent to $S > 0$.

When $S > 0$, invert the displayed factorization:

$$M^{-1} = \begin{pmatrix} 1 & 0 \\ -B^{-1} \gamma & I \end{pmatrix} \begin{pmatrix} S^{-1} & 0 \\ 0 & B^{-1} \end{pmatrix} \begin{pmatrix} 1 & -\gamma^T B^{-1} \\ 0 & I \end{pmatrix}.$$

Multiplying these three factors gives exactly the stated block formula. $\square$

Lemma 4 (Rooted branch estimate). *Let C be a finite weighted tree rooted at a vertex r . Let A be its Gram matrix, and assume A is positive definite. Suppose*

$$w(r) \geq d_C(r) + 1, \quad w(u) \geq d_C(u) \quad \text{for every } u \neq r.$$

Let $G = A^{-1}$, and define

$$\alpha = (A^{-1})_{rr} = e_r^T G e_r,$$

where e_r is the coordinate vector of r . Then $0 < \alpha \leq 1$, and for every $x \in L(C)$, if $c = x_r$, then for every $k \in \mathbb{Z}$,

$$(2k - 1)c - x^2 \leq k(k - 1)\alpha.$$

Moreover, if $k = 0$ or $k = 1$, and equality holds with $x \neq 0$, then C contains a vertex of weight 1.

Proof. We first prove a parity estimate for inverse Gram matrices. We claim that, under the hypotheses of the lemma,

$$0 < \alpha \leq 1$$

and that every vector $\lambda \in \mathbb{Z}^C$ satisfying

$$\lambda_r \equiv 1 \pmod{2}, \quad \lambda_u \equiv 0 \pmod{2} \quad (u \neq r)$$

also satisfies

$$\lambda^T G \lambda \geq \alpha.$$

This is proved by induction on $|C|$.

Reviewer 2

It would be nice to isolate this statement as a lemma, to break up a somewhat long proof.

If $C = \{r\}$, then $A = [w(r)]$. The root hypothesis gives $w(r) \geq 1$, and positive definiteness also gives $w(r) > 0$. Thus

$$\alpha = \frac{1}{w(r)} \leq 1.$$

If $\lambda_r = s$ is odd, then $s^2 \geq 1$, so

$$\lambda^T G \lambda = \frac{s^2}{w(r)} \geq \frac{1}{w(r)} = \alpha.$$

This proves the base case.

Assume now that $|C| > 1$. Let $C_1, \dots, C_m$ be the connected components of $C - \{r\}$, and let r_i be the unique vertex of C_i adjacent to r . Root C_i at r_i . Let A_i be the Gram matrix of C_i , set $G_i = A_i^{-1}$, let $e_i = e_{r_i}$, and put

$$\alpha_i = e_i^T G_i e_i.$$

Each A_i is positive definite because it is a principal submatrix of A . Also, each rooted tree C_i satisfies the same type of weight hypotheses: since $r_i \neq r$,

$$w(r_i) \geq d_C(r_i) = d_{C_i}(r_i) + 1,$$

and every other vertex $u \in C_i$ satisfies

$$w(u) \geq d_C(u) = d_{C_i}(u).$$

Therefore, by induction,

$$0 < \alpha_i \leq 1 \quad \text{for all } i.$$

Order the vertices of C by putting r first and then the vertices of $C_1, \dots, C_m$. With respect to this order,

$$A = \begin{pmatrix} w(r) & -\eta^T \\ -\eta & B \end{pmatrix}, \quad B = A_1 \oplus \dots \oplus A_m,$$

where the C_i -block of η is e_i . By Lemma 3, if

$$S = w(r) - \eta^T B^{-1} \eta,$$

then $S > 0$ and

$$\alpha = (A^{-1})_{rr} = \frac{1}{S}.$$

Since $B^{-1} = G_1 \oplus \dots \oplus G_m$, we have

$$\eta^T B^{-1} \eta = \sum_i \alpha_i.$$

Also $m = d_C(r)$, so the root hypothesis and the inductive bounds $\alpha_i \leq 1$ imply

$$S = w(r) - \sum_i \alpha_i \geq (m+1) - m = 1.$$

Thus $0 < \alpha = 1/S \leq 1$.

Now let $\lambda \in \mathbb{Z}^C$ have odd root coordinate and even nonroot coordinates. In the above ordering, write

$$\lambda = (s, 2\mu_1, \dots, 2\mu_m),$$

where s is odd and $\mu_i \in \mathbb{Z}^{C_i}$. Define

$$\rho_i = e_i^T G_i \mu_i, \quad E_i = \mu_i^T G_i \mu_i, \quad \rho = \sum_i \rho_i.$$

Using the inverse formula from Lemma 3 with $\gamma = -\eta$, we obtain

$$\lambda^T G \lambda = 4 \sum_i E_i + \frac{(s + 2\rho)^2}{S}.$$

Reviewer 2

I found I had to work this out on scratch paper, and it would have been convenient to say a little more here. (I left out a term and got confused.)

For each i , the vectors $2\mu_i - e_i$ and $2\mu_i + e_i$ have odd coordinate at the root r_i and even coordinates elsewhere in C_i . The induction hypothesis applied to C_i gives

$$(2\mu_i - e_i)^T G_i (2\mu_i - e_i) \geq \alpha_i, \quad (2\mu_i + e_i)^T G_i (2\mu_i + e_i) \geq \alpha_i.$$

Expanding these two inequalities gives

$$4E_i - 4\rho_i + \alpha_i \geq \alpha_i, \quad 4E_i + 4\rho_i + \alpha_i \geq \alpha_i.$$

Hence

$$E_i \geq |\rho_i|.$$

Therefore

$$\lambda^T G \lambda \geq 4 \sum_i |\rho_i| + \frac{(s + 2\rho)^2}{S} \geq 4|\rho| + \frac{(s + 2\rho)^2}{S}.$$

Set $y = 2\rho$. Then

$$\lambda^T G \lambda \geq 2|y| + \frac{(s + y)^2}{S}.$$

Since $S \geq 1$, it is enough to prove

$$(s + y)^2 + 2S|y| \geq 1.$$

Let $a = |s| \geq 1$ and $u = |y|$. If $sy \geq 0$, then $|s + y| = a + u \geq 1$, so the inequality is immediate. If $sy < 0$, then

$$(s + y)^2 + 2S|y| \geq (a - u)^2 + 2u = (u - (a - 1))^2 + 2a - 1 \geq 1.$$

Thus $S\lambda^T G \lambda \geq 1$, and hence

$$\lambda^T G \lambda \geq \frac{1}{S} = \alpha.$$

This completes the induction and proves the parity estimate.

We now derive the stated branch inequality. Fix $x \in L(C)$, let $c = x_r$, and set $Q = x^2 = x^T A x$. For $k \in \mathbb{Z}$, put

$$t = 2k - 1.$$

Define

$$\lambda = 2Ax - te_r.$$

Because A has integer entries and x has integer coordinates, $\lambda \in \mathbb{Z}^C$. Its root coordinate is even minus the odd integer t , hence is odd, while every nonroot coordinate is even. The parity estimate therefore gives

$$\lambda^T G \lambda \geq \alpha.$$

Using $G = A^{-1}$, we expand:

$$\lambda^T G \lambda = (2Ax - te_r)^T G (2Ax - te_r) = 4Q - 4tc + t^2\alpha.$$

Hence

$$4Q - 4tc + t^2\alpha \geq \alpha,$$

or equivalently

$$tc - Q \leq \frac{t^2 - 1}{4}\alpha.$$

Since $t = 2k - 1$,

$$\frac{t^2 - 1}{4} = \frac{(2k - 1)^2 - 1}{4} = k(k - 1).$$

Therefore

$$(2k - 1)c - x^2 \leq k(k - 1)\alpha.$$

It remains to prove the equality statement for $k = 0, 1$. For every $x \in L(C)$, expanding the form gives

$$x^2 = \sum_{\{u,v\} \in E(C)} (x_u - x_v)^2 + \sum_{u \in C} (w(u) - d_C(u))x_u^2.$$

Indeed, in the first sum, the term x_u^2 appears once for each edge incident to u , hence $d_C(u)$ times, and each edge contributes the cross term $-2x_u x_v$. Therefore the displayed expression equals

$$\sum_{u \in C} w(u)x_u^2 - 2 \sum_{\{u,v\} \in E(C)} x_u x_v = x^2.$$

By the hypotheses,

$$w(r) - d_C(r) \geq 1, \quad w(u) - d_C(u) \geq 0 \quad (u \neq r),$$

so

$$x^2 - c^2 = \sum_{\{u,v\} \in E(C)} (x_u - x_v)^2 + (w(r) - d_C(r) - 1)c^2 + \sum_{u \neq r} (w(u) - d_C(u))x_u^2 \geq 0.$$

Thus $x^2 \geq c^2$.

Suppose first that $k = 1$, equality holds in the branch inequality, and $x \neq 0$. Since $k(k - 1) = 0$, equality says

$$c - x^2 = 0.$$

Thus $x^2 = c$. Positive definiteness gives $x^2 > 0$, so $c > 0$, and hence $x^2 = |c|$. Since $x^2 \geq c^2$, we get $|c| \geq c^2$. Because $c \in \mathbb{Z}$ and $c \neq 0$, this forces $c = 1$, and then $x^2 = c^2 = 1$.

Similarly, if $k = 0$, equality says

$$-c - x^2 = 0.$$

Thus $x^2 = -c > 0$, so $c < 0$. Again $x^2 = |c|$, and the inequality $x^2 \geq c^2$ forces $c = -1$ and $x^2 = c^2 = 1$.

In either case, $|c| = 1$ and $x^2 = c^2$. Therefore equality holds in the nonnegative sum for $x^2 - c^2$, so every summand in that sum is zero. In particular, each edge term $(x_u - x_v)^2$ is zero. Since C is connected, all coordinates of x are equal to $c = \pm 1$.

If C has only one vertex, then $C = \{r\}$. The equality $x^2 = c^2$, with $c^2 = 1$, gives

$$(w(r) - d_C(r) - 1)c^2 = 0.$$

Since $d_C(r) = 0$, we obtain $w(r) = 1$.

If C has more than one vertex, choose a vertex $u \neq r$ at maximal distance from r . Such a vertex is a nonroot leaf, so $d_C(u) = 1$. Because all coordinates of x are ± 1 , we have $x_u^2 = 1$. The vanishing of the corresponding nonroot summand gives

$$(w(u) - d_C(u))x_u^2 = 0,$$

hence $w(u) - d_C(u) = 0$, and therefore $w(u) = 1$. Thus, whenever equality holds for $k = 0$ or $k = 1$ with $x \neq 0$, the branch C contains a vertex of weight 1. $\square$

Lemma 5 (Calculation at a vertex). *Let T be a finite weighted tree whose form is positive definite, and let b be a vertex of T . Let C range over the components of $T - \{b\}$. For each such component C , let r_C be the unique vertex of C adjacent to b , let A_C be the Gram matrix of C , and define*

$$\alpha_C = (A_C^{-1})_{r_C r_C}.$$

Then

$$\sum_C \alpha_C < w(b).$$

Moreover, if

$$x = kb + \sum_C x_C,$$

where $k \in \mathbb{Z}$ and $x_C \in L(C)$, and if $c_C = (x_C)_{r_C}$, then

$$x \cdot (b - x) = k(1 - k)w(b) + \sum_C \left((2k - 1)c_C - x_C^2 \right).$$

Proof. If $T = \{b\}$, then there are no components C , the sum $\sum_C \alpha_C$ is empty, and the first assertion says $0 < w(b)$, which follows from positive definiteness because $w(b) = b^2$. The displayed formula also reduces immediately to

$$(kb) \cdot (b - kb) = k(1 - k)w(b).$$

Thus assume $T \neq \{b\}$.

Order the vertices of T by putting b first, followed by the vertices in the components C of $T - \{b\}$. With respect to this order, the full Gram matrix has block form

$$\begin{pmatrix} w(b) & -\eta^T \\ -\eta & B \end{pmatrix}, \quad B = \bigoplus_C A_C,$$

where the C -block of η is e_{r_C} . The matrix B is positive definite because it is a principal submatrix of the positive definite Gram matrix of T . Applying Lemma 3, the Schur complement is positive:

$$w(b) - \eta^T B^{-1} \eta > 0.$$

Since $B^{-1} = \bigoplus_C A_C^{-1}$, we have

$$\eta^T B^{-1} \eta = \sum_C \alpha_C.$$

Therefore

$$\sum_C \alpha_C < w(b).$$

Now write

$$x = kb + \sum_C x_C.$$

Different components of $T - \{b\}$ are orthogonal to one another, and b pairs nontrivially with a component C only at the root r_C , where $b \cdot r_C = -1$. Hence

$$b \cdot x = kw(b) - \sum_C c_C.$$

Also,

$$x^2 = k^2 w(b) - 2k \sum_C c_C + \sum_C x_C^2.$$

Therefore

$$\begin{aligned} x \cdot (b - x) &= x \cdot b - x^2 \\ &= \left(kw(b) - \sum_C c_C \right) - \left(k^2 w(b) - 2k \sum_C c_C + \sum_C x_C^2 \right) \\ &= k(1 - k)w(b) + \sum_C \left((2k - 1)c_C - x_C^2 \right). \end{aligned}$$

This proves the formula. □

Proof of the theorem

Proof of Theorem 1. Let b denote the unique vertex of T satisfying $w(b) < d_T(b)$. Thus every vertex $u \neq b$ satisfies

$$w(u) \geq d_T(u).$$

If T contains a vertex of weight 1, then Lemma 2 shows that this vertex is irreducible, and the theorem follows. Hence assume from now on that T contains no vertex of weight 1. We will prove that b is irreducible.

Suppose, for contradiction, that b is reducible. Then there exists $x \in L(T)$ such that

$$x \neq 0, \quad b - x \neq 0, \quad x \cdot (b - x) \geq 0.$$

Decompose x according to the components of $T - \{b\}$:

$$x = kb + \sum_C x_C,$$

where $k \in \mathbb{Z}$, each $x_C \in L(C)$, and the sum ranges over the components C of $T - \{b\}$. Let r_C be the vertex of C adjacent to b , and put

$$c_C = (x_C)_{r_C}.$$

We now verify that each rooted component C , rooted at r_C , satisfies the hypotheses of Lemma 4. Since $r_C \neq b$, the uniqueness of b gives

$$w(r_C) \geq d_T(r_C) = d_C(r_C) + 1,$$

because the only edge incident to r_C that is not in C is the edge from r_C to b . If $u \in C$ and $u \neq r_C$, then no edge incident to u was removed when passing from T to C , so

$$w(u) \geq d_T(u) = d_C(u).$$

Finally, the Gram matrix A_C of C is positive definite because it is a principal submatrix of the positive definite Gram matrix of T . Thus Lemma 4 applies to each component C .

Let

$$\alpha_C = (A_C^{-1})_{r_C r_C}.$$

Applying Lemma 4 to each component, with the same integer k , gives

$$(2k - 1)c_C - x_C^2 \leq k(k - 1)\alpha_C.$$

Using Lemma 5, we obtain

$$\begin{aligned} x \cdot (b - x) &= k(1 - k)w(b) + \sum_C \left((2k - 1)c_C - x_C^2 \right) \\ &\leq k(1 - k)w(b) + k(k - 1) \sum_C \alpha_C \\ &= k(k - 1) \left(\sum_C \alpha_C - w(b) \right). \end{aligned}$$

By Lemma 5,

$$\sum_C \alpha_C - w(b) < 0.$$

If $k \notin \{0, 1\}$, then $k(k - 1) > 0$, so the last upper bound is strictly negative. This contradicts $x \cdot (b - x) \geq 0$. Therefore

$$k \in \{0, 1\}.$$

For $k = 0$ or $k = 1$, we have $k(1 - k) = 0$, and Lemma 4 gives

$$(2k - 1)c_C - x_C^2 \leq 0 \quad \text{for every component } C.$$

Lemma 5 therefore gives

$$x \cdot (b - x) = \sum_C \left((2k - 1)c_C - x_C^2 \right),$$

a sum of nonpositive terms. Since $x \cdot (b - x) \geq 0$, every term in this sum must be equal to 0. Hence, for every component C ,

$$(2k - 1)c_C - x_C^2 = 0.$$

If some $x_C \neq 0$, then the equality statement in Lemma 4 implies that C , and therefore T , contains a vertex of weight 1. This contradicts our standing assumption that T has no weight-one vertex. Thus

$$x_C = 0 \quad \text{for every component } C.$$

Consequently $x = kb$. If $k = 0$, then $x = 0$, contradicting $x \neq 0$. If $k = 1$, then $b - x = 0$, contradicting $b - x \neq 0$. Both possibilities are impossible.

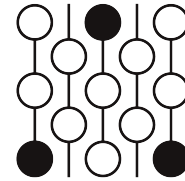
Therefore no such decomposition of b exists, and b is irreducible. Hence T contains an irreducible vertex. $\square$

Review of Submission 6B

First Proof Second Batch

Reviewer 3

June 4–8, 2026



1 Review

This gives a complete solution to the problem. The approach contains essentially all the same ideas as the known human proof. However, the proof here makes a direct beeline to the solution using elementary algebraic manipulation and so much of the motivation or interpretation of the steps is never explained. Although argument appears to be correct in all aspects, it seems likely that certain parts could be optimized.

Exposition In terms of the writing, I was not convinced by some of the notational and stylistic choices. For example writing x^2 for the self-pairing of a vector x is particularly jarring. Especially if one is also calling vectors p or some other variable name that is typically used for an integer. The choice of how much detail to provide is uneven: some parts are extremely terse whereas others are described in excessive detail. The level of writing here is more akin to a first draft than a finished paper. It is mathematically correct, but needs a lot of polishing.

Correctness. The solution is correct.

Novelty. The overall strategy of this solution very loosely follows that of the human solution. However any geometric or conceptual insight behind the steps is replaced by unmotivated algebraic manipulations. In general, the proof is extremely specialized to the case at hand and does not appear to contain any particular insights that could be directly generalized to related problems.

2 Recommendation

Based on the provided rubric, I would rate the solution to be essentially flawless. Although I think the exposition could be improved, it contains a complete solution in which all the details appear correct.

3 Submission with inline comments

Overview / Strategy

The proof has two main parts. First, a vertex of weight 1 is automatically irreducible, because a decomposition $u = p + q$ with $p \cdot q \geq 0$ would force $1 = u^2 \geq p^2 + q^2 \geq 2$. Thus, after this easy case, we may assume that no vertex has weight 1.

In that remaining case, let b be the unique vertex with $w(b) < d_T(b)$. We prove that b itself is irreducible. The key input is a rooted-branch estimate for each component of $T - \{b\}$. That estimate is proved by induction using a Schur-complement calculation and a parity argument for the inverse Gram matrix. Combining the branch estimates with the Schur complement at b , any hypothetical decomposition $b = x + (b - x)$ with $x \cdot (b - x) \geq 0$ is forced to have $x = 0$ or $x = b$, contradicting the requirement that both summands be nonzero.

Statement and notation

Let T be a finite tree with vertex set V , edge set E , and weight function $w : V \rightarrow \mathbb{Z}$. We write $d_T(v)$ for the degree of a vertex v in T . Let $L(T)$ be the free abelian group generated by V . We identify a vertex with the corresponding basis element of $L(T)$. The bilinear form on $L(T)$ is the symmetric form determined on generators by

$$u \cdot v = \begin{cases} w(u), & u = v, \\ -1, & \{u, v\} \in E, \\ 0, & u \neq v \text{ and } \{u, v\} \notin E. \end{cases}$$

For $x \in L(T)$, write $x^2 = x \cdot x$. The form is called positive definite if its extension to $L(T) \otimes_{\mathbb{Z}} \mathbb{R}$ is positive definite. Since the Gram matrix has integer entries, if the form is positive definite, then every nonzero $z \in L(T)$ satisfies $z^2 \in \mathbb{Z}_{>0}$.

A nonzero lattice element $y \in L(T)$ is called *irreducible* if there do not exist nonzero $a, c \in L(T)$ such that

$$y = a + c \quad \text{and} \quad a \cdot c \geq 0.$$

Theorem 1 (Original problem). *Let T be a finite weighted tree with vertex set V , edge set E , and weight function $w : V \rightarrow \mathbb{Z}$. Equip $L(T)$ with the symmetric bilinear form above. Suppose this form is positive definite, and suppose there is exactly one vertex $b \in V$ such that*

$$w(b) < d_T(b).$$

Then T contains a vertex u such that u is irreducible in $L(T)$.

For a connected subgraph or subtree C , write $d_C(u)$ for the degree of u inside C . The group $L(C)$ is the free abelian group on the vertices of C , equipped with the restricted form. If $x \in L(C)$, write x_u for the coefficient of the vertex u in x .

Auxiliary lemmas

Lemma 2 (A weight-one vertex is irreducible). *Assume the form on $L(T)$ is positive definite. If $u \in V$ satisfies $w(u) = 1$, then u is irreducible.*

Proof.

Reviewer 3

Writing p, q for vectors and p^2, q^2 for their pairing is an odd notational choice, since conventionally p, q are used to denote integers.

Suppose, for contradiction, that

$$u = p + q$$

where $p, q \in L(T)$ are nonzero and $p \cdot q \geq 0$. Since the form is positive definite and integral, $p^2, q^2 \in \mathbb{Z}_{>0}$. Hence

$$1 = u^2 = (p + q)^2 = p^2 + q^2 + 2p \cdot q \geq 1 + 1 + 0 = 2,$$

which is impossible. Therefore no such decomposition exists, and u is irreducible. $\square$

Lemma 3 (Schur complement calculation). *Let*

$$M = \begin{pmatrix} a & \gamma^T \\ \gamma & B \end{pmatrix},$$

where B is symmetric positive definite. Set

$$S = a - \gamma^T B^{-1} \gamma.$$

Then M is positive definite if and only if $S > 0$. When $S > 0$,

$$M^{-1} = \begin{pmatrix} S^{-1} & -S^{-1} \gamma^T B^{-1} \\ -B^{-1} \gamma S^{-1} & B^{-1} + B^{-1} \gamma S^{-1} \gamma^T B^{-1} \end{pmatrix}.$$

Proof. We have the factorization

$$M = \begin{pmatrix} 1 & \gamma^T B^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} S & 0 \\ 0 & B \end{pmatrix} \begin{pmatrix} 1 & 0 \\ B^{-1} \gamma & I \end{pmatrix}.$$

Indeed, multiplying the last two factors gives

$$\begin{pmatrix} S & 0 \\ \gamma & B \end{pmatrix},$$

and multiplying by the first factor gives

$$\begin{pmatrix} S + \gamma^T B^{-1} \gamma & \gamma^T \\ \gamma & B \end{pmatrix} = \begin{pmatrix} a & \gamma^T \\ \gamma & B \end{pmatrix}.$$

The first and third factors are transposes of one another and are invertible. Congruence by an invertible matrix preserves positive definiteness, so M is positive definite exactly when the middle block diagonal matrix is positive definite. Since B is already positive definite, this is equivalent to $S > 0$.

When $S > 0$, invert the displayed factorization:

$$M^{-1} = \begin{pmatrix} 1 & 0 \\ -B^{-1}\gamma & I \end{pmatrix} \begin{pmatrix} S^{-1} & 0 \\ 0 & B^{-1} \end{pmatrix} \begin{pmatrix} 1 & -\gamma^T B^{-1} \\ 0 & I \end{pmatrix}.$$

Multiplying these three factors gives exactly the stated block formula. □

Reviewer 3

The hypothesis that A be positive definite in the following lemma is redundant. It is implied by the condition on the weights.

Lemma 4 (Rooted branch estimate). *Let C be a finite weighted tree rooted at a vertex r . Let A be its Gram matrix, and assume A is positive definite. Suppose*

$$w(r) \geq d_C(r) + 1, \quad w(u) \geq d_C(u) \quad \text{for every } u \neq r.$$

Let $G = A^{-1}$, and define

$$\alpha = (A^{-1})_{rr} = e_r^T G e_r,$$

where e_r is the coordinate vector of r . Then $0 < \alpha \leq 1$, and for every $x \in L(C)$, if $c = x_r$, then for every $k \in \mathbb{Z}$,

$$(2k - 1)c - x^2 \leq k(k - 1)\alpha.$$

Moreover, if $k = 0$ or $k = 1$, and equality holds with $x \neq 0$, then C contains a vertex of weight 1.

Proof. We first prove a parity estimate for inverse Gram matrices. We claim that, under the hypotheses of the lemma,

$$0 < \alpha \leq 1$$

and that every vector $\lambda \in \mathbb{Z}^C$ satisfying

$$\lambda_r \equiv 1 \pmod{2}, \quad \lambda_u \equiv 0 \pmod{2} \quad (u \neq r)$$

also satisfies

$$\lambda^T G \lambda \geq \alpha.$$

This is proved by induction on $|C|$.

If $C = \{r\}$, then $A = [w(r)]$. The root hypothesis gives $w(r) \geq 1$, and positive definiteness also gives $w(r) > 0$.

Reviewer 3

Is there some reason why two explanations are given for the obvious fact that $w(r) \geq 1$?

Thus

$$\alpha = \frac{1}{w(r)} \leq 1.$$

If $\lambda_r = s$ is odd, then $s^2 \geq 1$, so

$$\lambda^T G \lambda = \frac{s^2}{w(r)} \geq \frac{1}{w(r)} = \alpha.$$

This proves the base case.

Assume now that $|C| > 1$. Let $C_1, \dots, C_m$ be the connected components of $C - \{r\}$, and let r_i be the unique vertex of C_i adjacent to r . Root C_i at r_i . Let A_i be the Gram matrix of C_i , set $G_i = A_i^{-1}$, let $e_i = e_{r_i}$, and put

$$\alpha_i = e_i^T G_i e_i.$$

Each A_i is positive definite because it is a principal submatrix of A . Also, each rooted tree C_i satisfies the same type of weight hypotheses: since $r_i \neq r$,

$$w(r_i) \geq d_C(r_i) = d_{C_i}(r_i) + 1,$$

and every other vertex $u \in C_i$ satisfies

$$w(u) \geq d_C(u) = d_{C_i}(u).$$

Therefore, by induction,

$$0 < \alpha_i \leq 1 \quad \text{for all } i.$$

Order the vertices of C by putting r first and then the vertices of $C_1, \dots, C_m$. With respect to this order,

$$A = \begin{pmatrix} w(r) & -\eta^T \\ -\eta & B \end{pmatrix}, \quad B = A_1 \oplus \dots \oplus A_m,$$

where the C_i -block of η is e_i . By Lemma 3, if

$$S = w(r) - \eta^T B^{-1} \eta,$$

then $S > 0$ and

$$\alpha = (A^{-1})_{rr} = \frac{1}{S}.$$

Since $B^{-1} = G_1 \oplus \dots \oplus G_m$, we have

$$\eta^T B^{-1} \eta = \sum_i \alpha_i.$$

Also $m = d_C(r)$, so the root hypothesis and the inductive bounds $\alpha_i \leq 1$ imply

$$S = w(r) - \sum_i \alpha_i \geq (m+1) - m = 1.$$

Thus $0 < \alpha = 1/S \leq 1$.

Now let $\lambda \in \mathbb{Z}^C$ have odd root coordinate and even nonroot coordinates. In the above ordering, write

$$\lambda = (s, 2\mu_1, \dots, 2\mu_m),$$

where s is odd and $\mu_i \in \mathbb{Z}^{C_i}$. Define

$$\rho_i = e_i^T G_i \mu_i, \quad E_i = \mu_i^T G_i \mu_i, \quad \rho = \sum_i \rho_i.$$

Using the inverse formula from Lemma 3 with $\gamma = -\eta$, we obtain

$$\lambda^T G \lambda = 4 \sum_i E_i + \frac{(s + 2\rho)^2}{S}.$$

For each i , the vectors $2\mu_i - e_i$ and $2\mu_i + e_i$ have odd coordinate at the root r_i and even coordinates elsewhere in C_i . The induction hypothesis applied to C_i gives

$$(2\mu_i - e_i)^T G_i (2\mu_i - e_i) \geq \alpha_i, \quad (2\mu_i + e_i)^T G_i (2\mu_i + e_i) \geq \alpha_i.$$

Expanding these two inequalities gives

$$4E_i - 4\rho_i + \alpha_i \geq \alpha_i, \quad 4E_i + 4\rho_i + \alpha_i \geq \alpha_i.$$

Hence

$$E_i \geq |\rho_i|.$$

Therefore

$$\lambda^T G \lambda \geq 4 \sum_i |\rho_i| + \frac{(s + 2\rho)^2}{S} \geq 4|\rho| + \frac{(s + 2\rho)^2}{S}.$$

Set $y = 2\rho$. Then

$$\lambda^T G \lambda \geq 2|y| + \frac{(s + y)^2}{S}.$$

Since $S \geq 1$, it is enough to prove

$$(s + y)^2 + 2S|y| \geq 1.$$

Let $a = |s| \geq 1$ and $u = |y|$. If $sy \geq 0$, then $|s + y| = a + u \geq 1$, so the inequality is immediate. If $sy < 0$, then

$$(s + y)^2 + 2S|y| \geq (a - u)^2 + 2u = (u - (a - 1))^2 + 2a - 1 \geq 1.$$

Thus $S\lambda^T G \lambda \geq 1$, and hence

$$\lambda^T G \lambda \geq \frac{1}{S} = \alpha.$$

This completes the induction and proves the parity estimate.

I would be inclined to pull out the parity estimate as a separate lemma.

We now derive the stated branch inequality. Fix $x \in L(C)$, let $c = x_r$, and set $Q = x^2 = x^T A x$. For $k \in \mathbb{Z}$, put

$$t = 2k - 1.$$

Define

$$\lambda = 2Ax - te_r.$$

Because A has integer entries and x has integer coordinates, $\lambda \in \mathbb{Z}^C$. Its root coordinate is even minus the odd integer t , hence is odd, while every nonroot coordinate is even. The parity estimate therefore gives

$$\lambda^T G \lambda \geq \alpha.$$

Using $G = A^{-1}$, we expand:

$$\lambda^T G \lambda = (2Ax - te_r)^T G (2Ax - te_r) = 4Q - 4tc + t^2\alpha.$$

Hence

$$4Q - 4tc + t^2\alpha \geq \alpha,$$

or equivalently

$$tc - Q \leq \frac{t^2 - 1}{4}\alpha.$$

Since $t = 2k - 1$,

$$\frac{t^2 - 1}{4} = \frac{(2k - 1)^2 - 1}{4} = k(k - 1).$$

Therefore

$$(2k - 1)c - x^2 \leq k(k - 1)\alpha.$$

It remains to prove the equality statement for $k = 0, 1$. For every $x \in L(C)$, expanding the form gives

$$x^2 = \sum_{\{u,v\} \in E(C)} (x_u - x_v)^2 + \sum_{u \in C} (w(u) - d_C(u))x_u^2.$$

Indeed, in the first sum, the term x_u^2 appears once for each edge incident to u , hence $d_C(u)$ times, and each edge contributes the cross term $-2x_u x_v$. Therefore the displayed expression equals

$$\sum_{u \in C} w(u)x_u^2 - 2 \sum_{\{u,v\} \in E(C)} x_u x_v = x^2.$$

By the hypotheses,

$$w(r) - d_C(r) \geq 1, \quad w(u) - d_C(u) \geq 0 \quad (u \neq r),$$

so

$$x^2 - c^2 = \sum_{\{u,v\} \in E(C)} (x_u - x_v)^2 + (w(r) - d_C(r) - 1)c^2 + \sum_{u \neq r} (w(u) - d_C(u))x_u^2 \geq 0.$$

Thus $x^2 \geq c^2$.

Suppose first that $k = 1$, equality holds in the branch inequality, and $x \neq 0$. Since $k(k-1) = 0$, equality says

$$c - x^2 = 0.$$

Thus $x^2 = c$. Positive definiteness gives $x^2 > 0$, so $c > 0$, and hence $x^2 = |c|$. Since $x^2 \geq c^2$, we get $|c| \geq c^2$. Because $c \in \mathbb{Z}$ and $c \neq 0$, this forces $c = 1$, and then $x^2 = c^2 = 1$.

Similarly, if $k = 0$, equality says

$$-c - x^2 = 0.$$

Thus $x^2 = -c > 0$, so $c < 0$. Again $x^2 = |c|$, and the inequality $x^2 \geq c^2$ forces $c = -1$ and $x^2 = c^2 = 1$.

In either case, $|c| = 1$ and $x^2 = c^2$. Therefore equality holds in the nonnegative sum for $x^2 - c^2$, so every summand in that sum is zero. In particular, each edge term $(x_u - x_v)^2$ is zero. Since C is connected, all coordinates of x are equal to $c = \pm 1$.

If C has only one vertex, then $C = \{r\}$. The equality $x^2 = c^2$, with $c^2 = 1$, gives

$$(w(r) - d_C(r) - 1)c^2 = 0.$$

Since $d_C(r) = 0$, we obtain $w(r) = 1$.

If C has more than one vertex, choose a vertex $u \neq r$ at maximal distance from r . Such a vertex is a nonroot leaf, so $d_C(u) = 1$. Because all coordinates of x are ± 1 , we have $x_u^2 = 1$. The vanishing of the corresponding nonroot summand gives

$$(w(u) - d_C(u))x_u^2 = 0,$$

hence $w(u) - d_C(u) = 0$, and therefore $w(u) = 1$. Thus, whenever equality holds for $k = 0$ or $k = 1$ with $x \neq 0$, the branch C contains a vertex of weight 1. $\square$

Lemma 5 (Calculation at a vertex). *Let T be a finite weighted tree whose form is positive definite, and let b be a vertex of T . Let C range over the components of $T - \{b\}$. For each such component C , let r_C be the unique vertex of C adjacent to b , let A_C be the Gram matrix of C , and define*

$$\alpha_C = (A_C^{-1})_{r_C r_C}.$$

Then

$$\sum_C \alpha_C < w(b).$$

Moreover, if

$$x = kb + \sum_C x_C,$$

where $k \in \mathbb{Z}$ and $x_C \in L(C)$, and if $c_C = (x_C)_{r_C}$, then

$$x \cdot (b - x) = k(1 - k)w(b) + \sum_C \left((2k - 1)c_C - x_C^2 \right).$$

Proof. If $T = \{b\}$, then there are no components C , the sum $\sum_C \alpha_C$ is empty, and the first assertion says $0 < w(b)$, which follows from positive definiteness because $w(b) = b^2$. The displayed formula also reduces immediately to

$$(kb) \cdot (b - kb) = k(1 - k)w(b).$$

Thus assume $T \neq \{b\}$.

Order the vertices of T by putting b first, followed by the vertices in the components C of $T - \{b\}$. With respect to this order, the full Gram matrix has block form

$$\begin{pmatrix} w(b) & -\eta^T \\ -\eta & B \end{pmatrix}, \quad B = \bigoplus_C A_C,$$

where the C -block of η is e_{r_C} . The matrix B is positive definite because it is a principal submatrix of the positive definite Gram matrix of T . Applying Lemma 3, the Schur complement is positive:

$$w(b) - \eta^T B^{-1} \eta > 0.$$

Since $B^{-1} = \bigoplus_C A_C^{-1}$, we have

$$\eta^T B^{-1} \eta = \sum_C \alpha_C.$$

Therefore

$$\sum_C \alpha_C < w(b).$$

Now write

$$x = kb + \sum_C x_C.$$

Different components of $T - \{b\}$ are orthogonal to one another, and b pairs nontrivially with a component C only at the root r_C , where $b \cdot r_C = -1$. Hence

$$b \cdot x = kw(b) - \sum_C c_C.$$

Also,

$$x^2 = k^2 w(b) - 2k \sum_C c_C + \sum_C x_C^2.$$

Therefore

$$\begin{aligned} x \cdot (b - x) &= x \cdot b - x^2 \\ &= \left(kw(b) - \sum_C c_C \right) - \left(k^2 w(b) - 2k \sum_C c_C + \sum_C x_C^2 \right) \\ &= k(1 - k)w(b) + \sum_C \left((2k - 1)c_C - x_C^2 \right). \end{aligned}$$

This proves the formula. □

Proof of the theorem

Proof of Theorem 1. Let b denote the unique vertex of T satisfying $w(b) < d_T(b)$. Thus every vertex $u \neq b$ satisfies

$$w(u) \geq d_T(u).$$

If T contains a vertex of weight 1, then Lemma 2 shows that this vertex is irreducible, and the theorem follows. Hence assume from now on that T contains no vertex of weight 1. We will prove that b is irreducible.

Suppose, for contradiction, that b is reducible. Then there exists $x \in L(T)$ such that

$$x \neq 0, \quad b - x \neq 0, \quad x \cdot (b - x) \geq 0.$$

Decompose x according to the components of $T - \{b\}$:

$$x = kb + \sum_C x_C,$$

where $k \in \mathbb{Z}$, each $x_C \in L(C)$, and the sum ranges over the components C of $T - \{b\}$. Let r_C be the vertex of C adjacent to b , and put

$$c_C = (x_C)_{r_C}.$$

We now verify that each rooted component C , rooted at r_C , satisfies the hypotheses of Lemma 4. Since $r_C \neq b$, the uniqueness of b gives

$$w(r_C) \geq d_T(r_C) = d_C(r_C) + 1,$$

because the only edge incident to r_C that is not in C is the edge from r_C to b . If $u \in C$ and $u \neq r_C$, then no edge incident to u was removed when passing from T to C , so

$$w(u) \geq d_T(u) = d_C(u).$$

Reviewer 3

This discussion of vertex degree seems like an unnecessary level of detail.

Finally, the Gram matrix A_C of C is positive definite because it is a principal submatrix of the positive definite Gram matrix of T . Thus Lemma 4 applies to each component C .

Let

$$\alpha_C = (A_C^{-1})_{r_C r_C}.$$

Applying Lemma 4 to each component, with the same integer k , gives

$$(2k - 1)c_C - x_C^2 \leq k(k - 1)\alpha_C.$$

Using Lemma 5, we obtain

$$\begin{aligned} x \cdot (b - x) &= k(1 - k)w(b) + \sum_C \left((2k - 1)c_C - x_C^2 \right) \\ &\leq k(1 - k)w(b) + k(k - 1) \sum_C \alpha_C \\ &= k(k - 1) \left(\sum_C \alpha_C - w(b) \right). \end{aligned}$$

By Lemma 5,

$$\sum_C \alpha_C - w(b) < 0.$$

If $k \notin \{0, 1\}$, then $k(k-1) > 0$, so the last upper bound is strictly negative. This contradicts $x \cdot (b-x) \geq 0$. Therefore

$$k \in \{0, 1\}.$$

For $k = 0$ or $k = 1$, we have $k(1-k) = 0$, and Lemma 4 gives

$$(2k-1)c_C - x_C^2 \leq 0 \quad \text{for every component } C.$$

Lemma 5 therefore gives

$$x \cdot (b-x) = \sum_C \left((2k-1)c_C - x_C^2 \right),$$

a sum of nonpositive terms. Since $x \cdot (b-x) \geq 0$, every term in this sum must be equal to 0. Hence, for every component C ,

$$(2k-1)c_C - x_C^2 = 0.$$

If some $x_C \neq 0$, then the equality statement in Lemma 4 implies that C , and therefore T , contains a vertex of weight 1. This contradicts our standing assumption that T has no weight-one vertex. Thus

$$x_C = 0 \quad \text{for every component } C.$$

Consequently $x = kb$. If $k = 0$, then $x = 0$, contradicting $x \neq 0$. If $k = 1$, then $b-x = 0$, contradicting $b-x \neq 0$. Both possibilities are impossible.

Therefore no such decomposition of b exists, and b is irreducible. Hence T contains an irreducible vertex. $\square$